

SULAMAN KHAN

AI Engineer | LLM Applications, RAG & Agentic Systems

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Full overlap with Gulf hours · 3–4 hours with UK/EU · US mornings covered

PROFESSIONAL SUMMARY

AI Engineer with 8 years of production software engineering, the last two focused on building LLM systems that run in production rather than in notebooks. I build Retrieval-Augmented Generation pipelines, LLM agents with tool calling, Model Context Protocol integrations, and structured-output extraction — with the multi-tenant isolation, evaluation, observability and cost discipline that production actually demands. That discipline comes from eight years on fintech platforms processing \$1M+ in transactions, systems serving 100,000+ daily users, and Layer 1 blockchain infrastructure. Fluent in Python, Go and Node.js. The AI stack is learnable in months; the engineering judgment took eight years, and I bring both.

CORE SKILLS

AI / LLM: Large Language Models, Retrieval-Augmented Generation (RAG), LLM agents, agentic systems, ReAct tool-calling loops, prompt engineering, structured outputs, embeddings, semantic search, RAG evaluation, Model Context Protocol (MCP), multi-tenant AI security

AI frameworks: LangChain, LangGraph, Anthropic Claude API, OpenAI API, Ollama, ChromaDB, pgvector, FastAPI, Server-Sent Events (SSE)

Computer vision: YOLO (object detection), OCR, ArcFace (face recognition), OpenCV

Backend: Python, Go, Node.js, TypeScript, Rust, PHP/Laravel, gRPC, microservices, REST APIs, concurrency

Data & cloud: PostgreSQL, pgvector, MySQL, Redis, Elasticsearch, GraphQL, AWS (Lambda, API Gateway), Docker, Kubernetes, GitHub Actions, Datadog

Practices: Evaluation harnesses, observability, guardrails, cost optimization, CI/CD, system design, data security

PROFESSIONAL EXPERIENCE

AI Engineer

Jan 2024 – Present

Vaival Technologies · Client: Vulcan Forged (US) · Remote

- **RAG pipelines:** Built end-to-end retrieval systems — document chunking, embedding generation, vector storage and retrieval across ChromaDB and pgvector — grounding LLM responses in source data, validated with an evaluation harness measuring retrieval hit-rate and answer faithfulness.
- **Multi-tenant AI security:** Implemented metadata-filtered retrieval enforcing strict tenant isolation, verified with security tests demonstrating zero cross-tenant data leakage — directly applicable to regulated fintech and banking environments.
- **LLM agents & MCP:** Developed agentic systems on the ReAct reasoning loop with step limits and input validation at the tool boundary, and built Model Context Protocol servers and clients exposing internal tools to any MCP-compatible client.
- **Production deployment:** Shipped RAG as an async FastAPI service with token streaming over Server-Sent Events, per-request cost and latency observability, provider-failure handling, and tenant isolation enforced at the API layer.
- **Reliability engineering:** Applied backend rigour to the AI layer — retry with exponential backoff, structured-output schema enforcement with defensive parsing, graceful degradation on malformed model output, and a multi-provider abstraction across Anthropic Claude and OpenAI.

Blockchain & Backend Engineer

Dec 2021 – Mar 2024

Vaival Technologies · Client: Vulcan Forged (US) · Remote

- **Backend at scale:** Built Go and Node.js services for a Layer 1 blockchain platform — block explorer, DEX, NFT marketplace, bridge and swap modules — serving real-time on-chain data over GraphQL. Optimized a related API to handle 1M+ requests.
- **Performance:** Migrated a Node.js service serving 1M users to Go, resolving persistent memory issues; raised throughput using goroutines and channels, with gRPC microservices for distributed communication.

- **Secure transactions:** Implemented transaction signing, DeFi protocol integration and secure smart-contract interaction — a security-first discipline that transfers directly to production AI systems.

Full Stack Engineer (Web & Mobile)

Jun 2018 – Dec 2021

B4USoft Pvt. Ltd. · Lahore, Pakistan

- **Fintech:** Built secure transaction, authentication and data-security features; the launched application processed \$1M+ in transactions in its first quarter.
- **High-traffic systems:** Designed and optimized a CMS supporting 100,000+ daily active users; built React Native e-commerce apps with Go APIs, Stripe and PayPal payments, and inventory management.
- **Cloud & delivery:** Serverless architecture on AWS Lambda with Docker-based deployment and Datadog monitoring — cut operational costs 40% while improving scalability. Led projects to CMMI 2.0 certification.

INDEPENDENT AI PROJECTS

Note on code visibility: production work for client engagements lives in client-owned repositories. Public architecture write-ups for the systems below are published at github.com/sulaman0.

- **Multi-Tenant RAG Service** — Production RAG API in Python, FastAPI, ChromaDB and Ollama with SSE token streaming, per-request tenant-isolated retrieval, cost and latency observability middleware, and provider-failure handling.
- **MCP Server & Client** — Both halves of a Model Context Protocol integration: a server exposing tools and resources, and a client that discovers and calls them at runtime, wired into a local-model agent loop.
- **Agentic System (ReAct)** — LLM agent with a think-act-observe tool-calling loop, step-limit safety caps and schema validation at the tool boundary; implemented in raw orchestration and rebuilt in LangGraph.
- **Looqer — AI Shopify Styling App:** AI styling app turning a short quiz into personalised outfit recommendations for Shopify merchants. Full monorepo: React and Polaris admin dashboard, Node/Express/Prisma API, vanilla-JS widget runtime embedded in storefronts, and a Shopify theme extension. Deployed on DigitalOcean with Neon PostgreSQL, Nginx and Cloudflare, with tiered subscription billing.
- **MyScan — Face-Based Health Monitoring:** Health-monitoring app estimating heart rate, heart-rate variability and breathing rate from a short facial video using remote photoplethysmography (ShenAI SDK). Integrated the SDK, built the camera and scanning flow, processed the returned metrics, and built the results and guidance experience. Designed for wellness, not diagnosis.
- **Support Ticket Triage Engine** — LLM system classifying priority, extracting structured fields and drafting replies from raw tickets, with graceful degradation and clean engine/interface separation.
- **Retail Shelf Detection** — Computer-vision pipeline combining YOLO for shelf and product detection, OCR for on-shelf text extraction, and ArcFace for face recognition into a single working system.
- **ZATCA Phase 2 E-Invoicing** — Saudi Arabia e-invoicing compliance (Fatoora, XML generation, certificate signing). Open-sourced; 13 stars and 7 forks, in use by other teams.

SELECTED ACHIEVEMENTS

- Fintech application processed \$1M+ in transactions within its first quarter post-launch.
- Migrated a 1M-user Node.js service to Go, eliminating recurring memory failures.
- Scaled a CMS platform to 100,000+ daily active users; optimized an API to 1M+ requests.
- Cut operational costs 40% through serverless architecture while improving scalability.
- Led projects to CMMI 2.0 certification; raised an app store rating from 4.3 to 4.9 and cut startup time to 10 seconds.
- Built and maintain three Shopify apps (Cartly, Sales Pop-up, Wheelify) used by thousands of merchants.

EDUCATION

B.S. Information Technology

2014 – 2018

University of the Punjab, Lahore, Pakistan

- Applied Concurrency in Go — goroutines, channels, concurrent systems design (2023)